

Se dă funcția de transfer

$$H(s) = \frac{2}{s+3}$$

Se cere discretizarea folosind metodele ZOH și Tustin, știind că $T = 0,5$ și $u(t) = \frac{1}{s}$.

① Metoda ZOH

$$\{t\} = \mathcal{Z}^{-1} \left\{ \frac{H(s)}{s} \right\}$$

pas 1: $\frac{H(s)}{s} = \frac{2}{s(s+3)}$

pas 2: Descumpunem în fracții simple

$$\frac{2}{s(s+3)} = \frac{\overset{s+3}{A}}{s} + \frac{\overset{s}{B}}{s+3}$$

$$2 = A(s+3) + sB$$

$$2 = As + 3A + Bs \Rightarrow$$

$$3A = 2 \Rightarrow A = \frac{2}{3}$$

și $As + Bs = 0 \Rightarrow \frac{2}{3} + B = 0 \Rightarrow B = -\frac{2}{3}$

$$\frac{H(s)}{s} = \frac{2}{3} \cdot \frac{1}{s} + \frac{-2}{3} \cdot \frac{1}{s+3}$$

pas 3: Calc. răsp. la semnal treaptă

$$\mathcal{Z}^{-1} \left\{ \frac{1}{s} \right\} = 1$$

$$\mathcal{Z}^{-1} \left\{ \frac{1}{s+a} \right\} = e^{-at}$$

$$\frac{1}{s+3} \Rightarrow \mathcal{L}^{-1} \left\{ \frac{1}{s+3} \right\} = e^{-3t}$$

$$f(t) = \frac{2}{3} - \frac{2}{3} \cdot e^{-3t}$$

pas 4: Esantionăm răsp. la semnalul treaptă-

$$f[k] = f(t) \Rightarrow f[k] = \frac{2}{3} - \frac{2}{3} \cdot e^{-3kT}$$

$$f[k] = \frac{2}{3} - \frac{2}{3} \cdot e^{-1,5k}$$

pas 5: Calculăm secv. impuls discretă cu ZOH

$$g[k] = f(k+1) - f(k)$$

$$g[k] = \left(\frac{2}{3} - \frac{2}{3} \cdot e^{-1,5(k+1)} \right) - \left(\frac{2}{3} - \frac{2}{3} \cdot e^{-1,5k} \right)$$

$$g[k] = \cancel{\frac{2}{3}} - \frac{2}{3} \cdot e^{-1,5(k+1)} - \cancel{\frac{2}{3}} + \frac{2}{3} \cdot e^{-1,5k}$$

$$g[k] = \frac{2}{3} \cdot e^{-1,5k} - \frac{2}{3} \cdot e^{-1,5(k+1)}$$

$$g[k] = \frac{2}{3} (e^{-1,5k} - e^{-1,5(k+1)})$$

$$g[k] = \frac{2}{3} (e^{-1,5k} - e^{-1,5k} \cdot e^{-1,5}) = \frac{2}{3} e^{-1,5k} (1 - e^{-1,5})$$

$$\text{Not. } a = e^{-1,5} \approx 0,2231$$

$$g[k] = \frac{2}{3} \cdot a^k (1-a)$$

pas 6: Stim că $a^k \xrightarrow{Z} \frac{1}{1-a \cdot z^{-1}}$

$$\Rightarrow H_{\text{dis}}(z) = \frac{2}{3} \cdot (1-a) \cdot \frac{1}{1-a \cdot z^{-1}}$$

Adăugăm apoi termenul liber

$$y(t) = B + u(t) + g[k]$$

$$B = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} \frac{2}{s+3} = 0$$

$$H(z) = \frac{2}{3} \cdot (1-a) \cdot \frac{1}{1-a \cdot z^{-1}} = \frac{2}{3} \cdot (1-0,2231) \cdot \frac{1}{1-0,2231 z^{-1}}$$

$$H(z) = \frac{2}{3} \cdot 0,7769 \cdot \frac{1}{1-0,2231 z^{-1}} = \frac{0,5179}{1-0,2231 z^{-1}}$$

② Metoda Tustin

$$\text{Substituația Tustin : } s = \frac{2}{T} \cdot \frac{z-1}{z+1}$$

$$\Rightarrow \text{vom înlocui } s \text{ în } \text{fct. de transfer cu } s = \frac{2}{0,5} \cdot \frac{z-1}{z+1}$$

$$H(z) = \frac{2}{\left(4 \cdot \frac{z-1}{z+1}\right) + 3}$$

$$\begin{aligned} \text{Calc. numitorul : } \left(4 \cdot \frac{z-1}{z+1}\right) + 3 &= \frac{4(z-1)}{z+1} + \frac{3}{1} = \frac{4z-4+3z+3}{z+1} \\ &= \frac{7z-1}{z+1} \end{aligned}$$

$$H(z) = \frac{2}{\frac{7z-1}{z+1}} = 2 \cdot \frac{z+1}{7z-1} = \frac{2z+2}{7z-1} \quad \left| \begin{array}{l} \text{(normalizare la} \\ \text{elem. } z) \\ \hline :z \end{array} \right.$$

$$H(z) = \frac{\frac{2}{7} z + \frac{2}{7}}{z - \frac{1}{7}} = \frac{0,2857z + 0,2857}{z - 0,1428}$$